

Notes: ‘n’ represents the dimension of the corresponding problem. ‘NI’ represents the total number of iterations. ‘NF’ represents the number of the function evaluation. ‘T’ represents the running time of the algorithm. ‘ $\|g^*\|$ ’ represents the gradient value for the solution. The values of ‘NI’, ‘NF’, ‘T’ and ‘ $\|g^*\|$ ’ are set as ‘NaN’ if and only if the number of iterations is greater than 3000.

Table 1: Numerical results for Test 2

Problem/n	SHCG-V1	SHCG-V3	CG-Descent	FhDY	MP+	SMABFGS
	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $	NI/NF/T/ $\ g^*\ $
dixmaana/3000	19/83/0.088/1.96e-07	21/81/0.056/9.15e-07	24/77/0.052/6.47e-07	19/83/0.058/7.02e-08	25/80/0.058/2.43e-07	45/96/0.069/1.69e-07
dixmaana/30000	17/86/0.469/3.56e-07	20/86/0.467/9.75e-07	31/88/0.472/8.56e-07	19/86/0.465/3.86e-07	28/87/0.478/3.87e-07	82/135/0.769/8.66e-07
dixmaana/600000	22/97/10.631/9.49e-07	14/84/9.216/6.03e-07	24/92/10.079/2.34e-07	17/90/9.839/6.14e-07	22/86/9.418/9.81e-07	92/142/15.734/3.12e-07
dixmaanb/6000	21/88/0.139/2.93e-07	18/84/0.105/6.07e-07	27/83/0.103/5.87e-07	16/82/0.102/2.89e-07	22/78/0.097/6.67e-07	49/101/0.126/5.24e-07
dixmaanb/15000	26/93/0.280/5.66e-07	20/87/0.262/8.58e-08	27/79/0.239/9.37e-07	22/90/0.273/4.55e-09	23/73/0.223/3.05e-07	72/125/0.401/3.71e-07
dixmaanb/60000	19/89/0.910/9.11e-07	23/93/0.950/6.05e-07	27/90/0.906/3.75e-07	23/98/0.984/9.52e-07	25/80/0.810/9.71e-07	75/131/1.345/3.00e-08
dixmaanc/3000	21/80/0.077/8.82e-07	18/81/0.057/9.54e-07	26/82/0.055/3.01e-07	19/84/0.057/5.05e-07	19/75/0.050/5.81e-08	40/86/0.055/2.22e-07
dixmaanc/150000	23/97/2.673/6.40e-07	21/92/2.512/9.06e-07	21/85/2.333/4.33e-07	20/95/2.600/6.21e-07	26/91/2.506/9.73e-07	90/146/4.075/6.97e-07
dixmaand/300000	15/86/4.698/3.48e-07	19/91/4.931/6.66e-07	27/93/5.048/9.15e-07	17/80/4.340/8.83e-07	28/93/5.075/1.66e-07	74/128/7.030/5.07e-07
dixmaane/1500	297/413/0.160/7.81e-07	315/399/0.129/6.24e-07	565/661/0.232/8.38e-07	308/589/0.200/9.87e-07	687/761/0.259/9.05e-07	382/430/0.141/5.60e-07
dixmaane/3000	431/601/0.386/9.98e-07	478/583/0.367/8.63e-07	793/911/0.586/9.52e-07	433/802/0.515/9.47e-07	761/851/0.555/7.66e-07	665/726/0.470/4.64e-07
dixmaane/15000	820/1054/3.177/8.37e-07	825/1039/3.166/7.91e-07	1907/2192/6.577/8.86e-07	753/1464/4.430/9.06e-07	1944/2037/6.277/9.42e-07	1443/1524/4.754/9.38e-07
dixmaanf/1500	270/391/0.154/9.23e-07	265/358/0.116/8.35e-07	433/486/0.170/9.08e-07	246/483/0.180/9.56e-07	600/690/0.226/9.78e-07	401/450/0.144/9.78e-07
dixmaanf/3000	290/409/0.254/7.05e-07	359/466/0.293/7.27e-07	534/608/0.378/9.03e-07	287/531/0.331/8.60e-07	787/890/0.564/9.34e-07	698/750/0.476/9.74e-07
dixmaanf/30000	838/1147/6.307/8.98e-07	783/996/5.480/6.94e-07	1804/1950/10.709/9.93e-07	834/1619/8.792/6.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaang/3000	128/203/0.034/8.32e-07	158/233/0.023/8.75e-07	196/248/0.025/7.09e-07	115/237/0.024/6.40e-07	181/240/0.024/5.21e-07	139/188/0.018/9.94e-07
dixmaang/1500	250/357/0.136/9.69e-07	332/432/0.147/8.96e-07	370/430/0.138/9.74e-07	266/520/0.166/9.22e-07	576/643/0.208/9.39e-07	327/381/0.126/7.25e-07
dixmaang/15000	1609/2151/6.548/4.16e-07	1648/1986/6.066/8.60e-07	1221/1314/3.999/8.31e-07	NaN/NaN/NaN/NaN	1995/2104/6.483/9.60e-07	1016/1079/3.349/7.24e-07
dixmaanh/15000	798/1089/3.426/6.72e-07	719/889/2.735/9.80e-07	1040/1167/3.546/9.25e-07	520/995/2.984/6.81e-07	1768/1856/5.692/9.04e-07	NaN/NaN/NaN/NaN
dixmaanh/21000	613/885/3.677/7.40e-07	1984/4380/17.894/8.67e-07	1436/1573/6.554/9.79e-07	851/1615/6.585/9.00e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanh/30000	874/1178/6.675/8.42e-07	1095/1358/7.737/8.46e-07	1432/1542/8.687/9.78e-07	921/1738/9.761/9.95e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaani/30	217/319/0.023/8.31e-07	203/266/0.007/9.81e-07	421/508/0.013/7.93e-07	202/398/0.012/9.14e-07	422/524/0.017/5.65e-07	257/305/0.009/8.33e-07
dixmaani/90	595/797/0.038/9.48e-07	632/746/0.033/8.92e-07	839/913/0.042/7.41e-07	477/904/0.041/8.24e-07	1632/1732/0.076/8.79e-07	948/1015/0.042/9.27e-07
dixmaani/210	1516/2041/0.135/7.77e-07	1242/1452/0.093/8.82e-07	NaN/NaN/NaN/NaN	1185/2267/0.140/9.99e-07	NaN/NaN/NaN/NaN	1870/1968/0.125/8.96e-07
dixmaanj/6000	858/1161/1.527/7.17e-07	745/925/1.141/6.73e-07	1481/1670/2.071/7.11e-07	592/1081/1.325/8.21e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaank/1500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	1653/3166/1.019/9.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanl/300	1113/1523/0.179/8.07e-07	1305/1572/0.156/7.68e-07	NaN/NaN/NaN/NaN	868/1647/0.133/9.20e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dixmaanl/21000	910/1234/5.079/6.05e-07	952/1181/4.886/9.08e-07	1282/1467/6.029/8.08e-07	1331/2530/10.179/9.23e-07	1241/1339/5.424/7.61e-07	NaN/NaN/NaN/NaN
dixmaanl/30000	831/1157/6.306/9.53e-07	1141/1394/7.643/9.84e-07	1410/1570/8.632/8.03e-07	674/1299/7.056/8.20e-07	1913/2065/11.487/9.68e-07	NaN/NaN/NaN/NaN
dixon3dq/50	524/735/0.021/5.65e-07	488/623/0.006/9.40e-07	776/862/0.009/7.99e-07	616/1166/0.013/8.84e-07	NaN/NaN/NaN/NaN	493/537/0.005/8.84e-07
dixon3dq/100	936/1260/0.013/9.01e-07	936/1144/0.012/8.27e-07	NaN/NaN/NaN/NaN	821/1576/0.015/9.16e-07	NaN/NaN/NaN/NaN	1158/1235/0.011/9.27e-07
dixon3dq/150	1785/2406/0.022/9.68e-07	1815/2184/0.021/8.29e-07	NaN/NaN/NaN/NaN	1223/2322/0.021/5.56e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
dqdrtic/10000	95/226/0.026/6.95e-07	77/228/0.016/5.19e-07	152/258/0.025/3.02e-07	74/222/0.020/9.40e-07	312/457/0.050/7.72e-07	142/209/0.024/2.65e-07
dqdrtic/50000	80/206/0.066/7.68e-07	100/294/0.089/7.26e-07	150/265/0.094/8.12e-07	62/210/0.070/5.14e-07	336/531/0.195/8.33e-07	193/263/0.114/3.59e-07
dqdrtic/150000	95/232/0.282/7.57e-07	76/221/0.262/3.47e-07	149/263/0.336/3.43e-07	86/241/0.279/6.97e-07	296/436/0.701/9.59e-07	138/213/0.351/7.72e-07
dqrtic/5000	54/164/0.174/3.14e-07	56/168/0.169/6.27e-07	62/157/0.158/4.63e-07	41/156/0.166/4.19e-07	187/1226/1.232/9.99e-07	96/185/0.194/3.43e-07
edensch/500	30/105/0.018/2.61e-07	44/195/0.019/8.63e-07	74/399/0.040/9.74e-07	68/474/0.052/3.98e-07	31/84/0.009/9.85e-07	144/941/0.092/9.60e-07
edensch/20000	67/385/1.508/7.93e-07	78/502/1.966/4.01e-07	120/849/3.300/9.79e-07	51/269/1.052/5.85e-07	45/213/0.838/5.72e-07	394/2923/11.440/9.95e-07
edensch/120000	210/1663/35.983/8.37e-07	69/375/8.182/6.99e-07	111/827/17.941/8.11e-07	125/965/20.848/2.12e-07	58/308/6.689/9.15e-07	882/7254/156.739/9.96e-07
eg2/100	236/752/0.027/8.64e-07	303/1422/0.019/7.68e-07	1057/6524/0.083/9.94e-07	538/1585/0.023/4.46e-07	NaN/NaN/NaN/NaN	1086/1250/0.018/8.55e-07
eg2/200	NaN/NaN/NaN/NaN	343/1322/0.028/8.24e-07	NaN/NaN/NaN/NaN	1122/5148/0.101/8.30e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
eg2/500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
fletcher/7000	139/1147/0.079/4.75e-07	260/2188/0.123/7.02e-07	179/1292/0.067/9.69e-07	391/3583/0.202/9.27e-07	475/4542/0.258/9.60e-07	NaN/NaN/NaN/NaN
freuroth/50	218/1174/0.020/9.11e-07	181/550/0.007/8.05e-07	383/2205/0.027/8.64e-07	236/1585/0.017/9.83e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
freuroth/500	NaN/NaN/NaN/NaN	1087/10723/0.146/9.94e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
freuroth/5000	NaN/NaN/NaN/NaN	923/9512/0.649/7.43e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
genrose/10000	234/392/0.050/5.52e-07	290/462/0.051/9.99e-07	302/418/0.050/8.30e-07	225/473/0.045/7.75e-07	752/877/0.095/9.80e-07	397/475/0.068/5.53e-07
himmelbg/1000	2/13/0.005/7.55e-30	2/13/0.001/7.55e-30	6/24/0.001/2.61e-08	2/13/0.001/7.57e-30	9/17/0.001/1.49e-18	8/12/0.001/1.59e-42
himmelbg/70000	2/15/0.033/8.37e-43	2/15/0.029/8.75e-43	6/26/0.037/5.05e-200	2/15/0.029/8.41e-43	11/42/0.061/0.00e+00	8/15/0.031/9.38e-10
himmelbg/600000	3/16/0.259/5.32e-26	3/16/0.255/1.30e-26	6/37/0.480/1.76e-30	3/16/0.261/8.54e-27	5/17/0.287/4.37e-12	11/19/0.343/8.27e-20
liarwhd/10000	233/680/0.072/3.52e-07	128/456/0.046/3.94e-07	411/823/0.060/3.49e-07	176/628/0.033/5.48e-07	NaN/NaN/NaN/NaN	853/1117/0.128/6.11e-07
liarwhd/30000	143/537/0.105/7.87e-07	164/562/0.107/5.06e-07	459/951/0.188/5.53e-08	201/732/0.135/2.54e-07	NaN/NaN/NaN/NaN	716/994/0.273/2.91e-07
liarwhd/120000	305/882/0.516/3.70e-07	164/651/0.372/1.51e-07	391/836/0.502/5.75e-07	315/1075/0.625/4.67e-07	NaN/NaN/NaN/NaN	679/1018/0.790/8.45e-07
penalty1/100	18/108/0.018/6.16e-07	19/109/0.002/8.13e-07	14/66/0.001/6.61e-07	18/107/0.002/3.82e-07	14/66/0.001/6.62e-07	21/68/0.002/4.28e-07
penalty1/10000	20/94/3.234/2.94e-09	20/92/3.146/4.52e-07	65/281/9.560/4.56e-07	20/91/3.095/8.17e-07	50/259/8.864/6.09e-07	113/180/6.205/3.87e-07
quartc/5000	54/164/0.171/3.14e-07	56/168/0.170/6.27e-07	62/157/0.160/4.63e-07	41/156/0.164/4.19e-07	187/1226/1.260/9.99e-07	96/185/0.203/3.43e-07
tridia/100	361/489/0.010/9.50e-07	251/341/0.004/8.82e-07	809/940/0.009/9.75e-07	353/664/0.007/6.72e-07	1305/1502/0.014/9.35e-07	365/426/0.004/4.46e-07
tridia/300	591/793/0.008/8.11e-07	623/768/0.008/9.58e-07	1702/1959/0.020/9.71e-07	789/1519/0.016/6.12e-07	NaN/NaN/NaN/NaN	797/871/0.009/9.36e-07
tridia/1500	1675/2260/0.036/9.30e-07	1960/2355/0.039/9.73e-07	NaN/NaN/NaN/NaN	1544/2944/0.046/6.80e-07	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
woods/5000	196/534/0.043/7.03e-07	158/434/0.017/8.84e-07	347/476/0.020/6.25e-07	263/640/0.025/7.03e-07	698/1918/0.077/9.63e-07	262/336/0.027/3.07e-07
woods/300000	230/578/1.482/2.66e-07	210/530/1.365/8.38e-07	461/613/1.805/6.53e-07	376/910/2.294/8.20e-07	938/2541/7.086/9.81e-07	324/401/1.511/8.73e-07
woods/1000000	179/436/3.699/4.15e-07	218/648/4.698/5.72e-07	341/493/3.941/9.18e-07	281/702/5.025/7.37e-07	991/2608/21.582/8.60e-07	396/479/5.473/6.24e-07
bdexp/1000	2/11/0.009/2.98e-133	2/11/0.001/2.99e-133	6/25/0.002/1.76e-16	2/11/0.001/3.25e-133	NaN/NaN/NaN/NaN	13/14/0.001/1.41e-11
bdexp/10000	2/17/0.011/2.62e-61	2/17/0.010/2.62e-61	7/21/0.010/7.59e-161	2/17/0.010/2.63e-61	NaN/NaN/NaN/NaN	37/39/0.031/0.00e+00
bdexp/100000	2/12/0.083/4.57e-106	2/12/0.080/4.57e-106	6/22/0.111/5.22e-44	2/12/0.081/4.57e-106	6/22/0.111/8.46e-38	30/37/0.247/0.00e+00
exdenschf/10000	38/114/0.029/5.03e-07	40/113/0.010/8.83e-07	25/94/0.008/7.73e-07	25/99/0.008/4.94e-08	27/95/0.008/8.85e-07	55/118/0.016/4.22e-07
exdenschb/3000	25/89/0.007/5.69e-07	26/89/0.003/9.86e-07	28/82/0.003/6.56e-07	19/84/0.003/8.43e-09	25/76/0.002/2.42e-07	50/98/0.004/8.07e-07
exdenschb/5000000	40/115/4.204/7.22e-07	49/125/4.388/3.21e-07	34/106/3.468/3.53e-07	32/107/3.418/4.32e-07	22/91/3.128/2.64e-07	104/164/7.878/4.03e-07
genquartic/50000	18/87/0.057/5.78e-07	25/93/0.035/3.19e-07	27/89/0.034/1.78e-08	24/116/0.046/3.82e-07	23/85/0.038/5.06e-07	63/117/0.062/9.80e-07
genquartic/100000	19/81/0.063/5.37e-07	21/86/0.057/7.90e-07	32/95/0.063/9.65e-07	19/96/0.063/3.50e-07	30/96/0.064/5.54e-07	75/128/0.101/3.92e-07
biggsb1/250	1282/1688/0.025/7.82e-07	1518/1838/0.021/9.91e-07	1912/2019/0.023/7.25e-07	1362/2593/0.026/9.63e-07	NaN/NaN/NaN/NaN	1737/1820/0.019/6.18e-07
biggsb1/500	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN
fletcbv3/100	873/1724/0.066/9.53e-07	986/2159/0.076/7.87e-07	1037/1324/0.041/5.49e-07	1472/2974/0.095/2.43e-07	NaN/NaN/NaN/NaN	648/833/0.026/8.72e-07
nonscomp/10000	NaN/NaN/NaN/NaN	NaN/NaN/NaN/NaN	56/122/0.009/8			